I. Project: Inviscid Flow with Random Data

The flow of an inviscid, compressible fluid from time 0 to time T in a channel of length L can be described by the one-dimensional compressible Euler equations:

$$\begin{aligned}\partial_t \varrho + \partial_x m &= 0, \\ \partial_t m + \partial_x (um) + \partial_x p &= 0, \\ \partial_t E + \partial_x ((E+p)u) &= 0. \end{aligned} \tag{1}$$

A solution is a tuple $(\varrho, m, E)\colon [0,T] \times [0,L] \to \mathbb{R}^3$ that solves the system (1) (in an appropriate sense).

Here, we denote:

- $\varrho(t,x) \geq 0$: the mass density of the fluid at time t and position x,
- $m(t,x)$: the momentum density of the fluid at time t and position x,
- $E(t,x) \geq 0$: the energy density at time t and position x,
- $u(t,x) = m(t,x)/\varrho(t,x)$: the velocity at time t and position x,
- $p(t,x) = (\gamma-1)(E(t,x) - \varrho(t,x)\frac{u(t,x)^2}{2})$: the pressure at time t and position x,
- $\gamma \in (1, 5/3)$: the adiabatic coefficient.

In the formulation we consider, the state of the fluid is determined by the *state variables* (ϱ, m, E). These are conserved quantities particularly well-suited for numerical discretization. However, initial and boundary values are often prescribed for (ϱ, u, p), as these quantities are physically easier to measure and some scenarios can be more compactly described in terms of them.

1 Deterministic Model

1. Briefly discuss the derivation of the Euler equations. What physical assumptions and laws underlie them? What mathematical regularity is assumed for the functions involved?

2. What solution concepts exist for system (1)? What is their relevance? Is the Cauchy problem on $\mathbb{R}_+ \times \mathbb{R}$ well-posed for the Euler equations under any solution concept? Also briefly mention some relevant results for the Euler equations in $d = 2, 3$ spatial dimensions.

3. Which numerical methods are suitable for the Euler equations? How can initial and boundary conditions be incorporated? Propose a suitable discretization. Ensure that in the case of periodic boundary conditions, total mass and energy are conserved. What should be considered regarding the time step size?

4. Implement your method. Test your method on the following problems. Numerically investigate the order of convergence of your method.

   (i) Periodic boundary conditions, $T = 0.5$, $L = 1$, $\gamma = 1.2$, and initial data

   $$(\varrho, u, p)(0, x) = (1 + \exp(-20(x - L/2)^2),\ 1,\ 10) \qquad \text{for all } x \in [0, L]$$

   (ii) Neumann boundary conditions, $T = 0.2$, $L = 1$, $\gamma = 1.4$, and initial data

   $$(\varrho, u, p)(0, x) = \begin{cases} (1,\ 0,\ 1) & \text{for all } x \in [0, L/2), \\ (0.125,\ 0,\ 0.1) & \text{for all } x \in [L/2, L]. \end{cases}$$

2 Model with Random Data

We now consider the case where initial data, boundary data, and γ are given as random variables. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space. Here, we restrict ourselves to the case $\Omega = (0, 1)$ with $\mathcal{A} = \mathcal{B}(\Omega)$, the Borel σ-algebra on $(0, 1)$. Thus, we assume that uncertainties in the data can be parameterized by their dependence on $\omega \in (0, 1)$.

1. Formulate a stochastic collocation method for this initial-boundary value problem. Choose as ansatz spaces with respect to Ω: piecewise constant functions, cubic splines, and polynomials. Implement your method. What form of convergence do you expect?

2. First, assume only the initial data depend on ω. Numerically investigate the following scenarios using the different ansatz spaces with respect to Ω:

   (i) Periodic boundary conditions, $T = 0.5$, $L = 1$, $\gamma = 1.2$, and initial data

   $$(\varrho, u, p)(0, x, \omega) = (1 + \exp(-20(x - L\omega)^2),\ 1,\ 10) \qquad \text{for all } x \in [0, L]$$

   (ii) Neumann boundary conditions, $T = 0.2$, $L = 1$, $\gamma = 1.4$, and initial data

   $$(\varrho, u, p)(0, x, \omega) = \begin{cases} (1,\ 0,\ 1) & \text{for all } x \in [0, \omega), \\ (0.125,\ 0,\ 0.1) & \text{for all } x \in [\omega, L]. \end{cases}$$

   Examine the behavior of your solutions with varying numbers of collocation points. Which discretization with respect to Ω is the most stable? Explain your observations.

3. Consider again the same initial and boundary data independent of ω as in Section 1.4, but choose the following $\gamma\colon \Omega \to \mathbb{R}$:

$$\gamma(\omega) = 1.1 + 0.5\omega.$$

Experimentally test the convergence of your method with respect to the number of collocation points in various L^p norms. Consider at least the probability measure $\mathbb{P}$ induced by the Borel-Lebesgue measure on Ω.

4. Now, we allow the boundary data to depend on ω. Consider $T = L = 1$, $\gamma = 1.4$, and the following initial data:

$$(\varrho, u, p)(0, x, \omega) = \begin{cases} (1,\ 2,\ 1) & \text{for all } x \in [0, L/2), \\ (0.7,\ 1,\ 0.7) & \text{for all } x \in [L/2, L]. \end{cases}$$

Additionally, let the boundary conditions be given as follows:

$$\begin{array}{ll} \text{Left boundary:} & \varrho(t,0,\omega) = \varrho_L(t,\omega), \quad p(t,0,\omega) = p_L(t,\omega), \quad u(t,0,\omega) = 2 \\ \text{Right boundary:} & \varrho_x(t,L,\omega) = m_x(t,L,\omega) = E_x(t,L,\omega) = 0, \end{array}$$

Consider the following cases:

(i) $\varrho_L(t,\omega) = 1 + \exp(-3\omega t)$, $p_L(t,\omega) = 1 + \exp(-3\omega t)$

(ii) $\varrho_L(t,\omega) = 1$, $p_L(t,\omega) = 1 + \exp(-3\omega t)$

Write a complete report in which you formulate and analyze the mathematical model, describe the numerical methods, and present the results of numerical simulations.

Literatur

[1] A. Bressan: Hyperbolic systems of conservation laws: the one dimensional Cauchy problem. OUP Oxford (2000).

[2] A. Chertock, M. Herty, A.S. Iskhakov, S. Janajra, A. Kurganov, M. Lukáčová-Medvid'ová: New High-Order Numerical Methods for Hyperbolic Systems of Nonlinear PDEs with Uncertainties. Commun. Appl. Math. Comput. 6 (2024), pp. 2011-2044.

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